

CS324 Assignment 2: CNN & RNN

This assignment implements three deep learning models using PyTorch: Multilayer Perceptron (MLP), Convolutional Neural Network (CNN), and Recurrent Neural Network (RNN).

Environment Setup

Prerequisites

- Python 3.7+
- pip

Installation

```
1 # Install PyTorch (CPU version)
2 pip install torch torchvision
3
4 # Install other dependencies
5 pip install numpy matplotlib scikit-learn jupyter notebook
```

For GPU support, install PyTorch with CUDA from pytorch.org.

Project Structure

```
1 Assignment2 - CNN&RNN/
2 |— Part 1/                               # MLP Implementation
3 |   |— pytorch_mlp.py                   # PyTorch MLP model
4 |   |— cifar10_mlp.py                   # CIFAR-10 MLP model
5 |   |— pytorch_train_mlp.py             # Training script
6 |   |— task2_compare_mlp.ipynb          # NumPy vs PyTorch comparison
7 |   |— task3_cifar10.ipynb              # CIFAR-10 classification
8 |   |— data/                           # CIFAR-10 dataset (auto-downloaded)
9 |
10 |— Part 2/                               # CNN Implementation
11 |   |— cnn_model.py                     # CNN architecture
12 |   |— cnn_train.py                     # Training script
13 |   |— cnn_analysis.ipynb               # Performance analysis
14 |
15 |— Part 3/                               # RNN Implementation
16 |   |— vanilla_rnn.py                   # Vanilla RNN from scratch
17 |   |— train.py                         # Training script
18 |   |— dataset.py                       # Palindrome dataset
19 |   |— palindrome_rnn_analysis.ipynb    # Length analysis
20 |
21 |— images/                               # Figures for report
22 |— report.tex                           # LaTeX report
23 |— compile.py                           # LaTeX compilation script
24 |— README.md                            # This file
```

Usage

Part 1: MLP

Task 2 - Compare NumPy and PyTorch:

```
1 | cd Part\ 1
2 | jupyter notebook task2_compare_mlp.ipynb
```

Task 3 - CIFAR-10 Classification:

```
1 | cd Part\ 1
2 | jupyter notebook task3_cifar10.ipynb
```

Part 2: CNN

Train CNN:

```
1 | cd Part\ 2
2 | python cnn_train.py --early_stopping --patience 5 --min_delta 0.05
```

Analysis:

```
1 | cd Part\ 2
2 | jupyter notebook cnn_analysis.ipynb
```

Part 3: RNN

Train RNN:

```
1 | cd Part\ 3
2 | python train.py --input_length 10 --num_hidden 128 --train_steps 500
```

Analysis:

```
1 | cd Part\ 3
2 | jupyter notebook palindrome_rnn_analysis.ipynb
```

Compile Report

```
1 | python compile.py
```

This generates `report.pdf` from `report.tex`.

Key Features

- **Part 1:** MLP implementation with NumPy/PyTorch comparison and CIFAR-10 classification
- **Part 2:** VGG-style CNN with batch normalization and early stopping
- **Part 3:** Vanilla RNN from scratch (no built-in modules) for palindrome prediction

Notes

- CIFAR-10 dataset will be automatically downloaded to `Part 1/data/` on first run
- All models use default hyperparameters as specified in the assignment
- Training logs and checkpoints are saved during training